A Comparative Backtesting Study of Machine Learning and Rule-Based Trading Strategies in Cryptocurrency Markets

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GitHub Repository: https://github.com/xdewu0508/QF4211Project

1 Introduction

Cryptocurrency markets trade continuously and lack conventional valuation anchors such as cash flows or earnings. As a result, market participants rely heavily on price action, volume, and technical indicators, while machine-learning models promise greater flexibility in capturing non-linear and sequential structure. The practical issue, however, is not whether a model can generate directional predictions in isolation, but whether those signals survive a realistic backtest once transaction costs and portfolio turnover are taken seriously.

This study therefore asks the following question: To what extent do ML-based trading strategies outperform traditional rule-based strategies and passive buy-and-hold benchmarks in terms of the Sharpe Ratio, once transaction costs and backtesting constraints are applied to major assets such as Bitcoin (BTC), Ethereum (ETH), Ripple (XRP), and Solana (SOL)? We evaluate four machine-learning models (Logistic Regression, XGBoost, LightGBM, and GRU), two traditional rule-based strategies (SMA and RSI), and a buy-and-hold benchmark under long-only, short-only, and long-short structures.

Our contribution is a cost-aware, cross-asset comparison with an additional fixed-window robustness check rather than a claim that machine learning consistently dominates cryptocurrency trading. The main analysis uses the canonical 2021–2025 sample, and the report later compares it with a 2020–

2024 rerun that keeps the same pipeline, feature construction, split ratios, models, thresholding rules, and backtesting assumptions. Across both windows, long-only strategies are the most stable active structure and machine-learning models provide pockets of improvement in selected settings, yet they do not deliver robust, statistically significant outperformance over simpler active rules or passive buy-and-hold within the current evaluation design.

2 Literature Review

Because cryptocurrency assets do not generate cash flows or dividends, technical indicators remain a common decision framework. Hudson and Urquhart (2021) show that technical rules can outperform passive exposure in selected cryptocurrency settings, but Liu et al. (2024) note that indicators such as RSI do not always behave in crypto the way they do in traditional asset classes. Transaction-cost assumptions are also pivotal: Svogun and Bazán-Palomino (2022) show that fees can materially change profitability assessments, while studies such as Ratia (2023) and Anghel (2021) focus mainly on long-only designs. The literature therefore leaves open how well rule-based trading survives realistic costs across richer position structures.

Against that backdrop, machine-learning approaches have become increasingly common in cryptocurrency forecasting, with linear classifiers, boosted trees, and recurrent networks all used to capture non-linear and temporal structure. Chatzis et al. (2025) report strong results for models such as XGBoost and recurrent networks, but those gains are often studied in more favorable settings than a fee-aware backtest. Belalova et al. (2023) argue that GRU can outperform LSTM when data are limited because its simpler gating structure reduces overfitting risk, while Jabbar and Jalil (2024) emphasize that apparent predictive gains often weaken sharply out of sample. That tension between modelling flexibility and trading robustness motivates our model set.

Against this literature, our project sits between replication and extension and is anchored mainly in Goutte et al. (2023), *Deep learning and technical analysis in cryptocurrency market.* We retain its hourly, feature-based framing, but compare ML models, rule-based strategies, and buy-and-hold across four assets and three strategy structures under a common `0.035%` transaction fee per unit

position change. The study prioritizes economic evaluation rather than raw classification quality, and the report also adds a shifted 2020–2024 rerun to test sensitivity to the evaluation window while keeping the substantive claims deliberately scoped to the submitted design.

3 Data Description and Preprocessing

3.1 Data Source and Assets

The dataset is drawn from Binance public hourly candlestick files for four liquid USDT spot pairs: BTC/USDT, ETH/USDT, XRP/USDT, and SOL/USDT. These assets were selected to provide a mix of large-cap and higher-beta behavior while keeping the market venue, fee assumptions, and data structure consistent across the analysis. BTC and ETH represent larger-cap assets with deeper institutional participation, while XRP and SOL add higher-beta and more regime-sensitive behavior.

3.2 Sample Period

The canonical raw hourly files span January 2021 to December 2025. After forward-return alignment, the main notebook works with `43,808` usable hourly observations per asset. This window covers the 2021 bull market, the 2022 drawdown, the 2023 recovery, the 2024 ETF-driven institutional surge, and the volatile conditions late in 2025. To test whether the later sample materially changes the conclusions, we also rerun the same workflow on a shifted 2020–2024 window in a sandbox environment and compare the results in Section 5 and Appendix C. That rerun strengthens the regime-sensitivity discussion, but it remains a fixed-window comparison rather than full walk-forward validation.

Hourly data were retained to remain aligned with the reference paper and to preserve an intraday decision horizon, while still being coarse enough to avoid the extreme noise and execution burden of sub-hourly strategies. Even so, hourly signals remain noisy and turnover-sensitive, which is important when interpreting the final results.

3.3 Data Construction

We construct four feature sets. `ohlc_raw` uses open, high, low, close, volume, trade count, and cyclical hour terms. `candle_raw` replaces the raw OHLC block with candlestick body and shadow representations while retaining volume, trade count, and cyclical terms. `ohlc_extended` and `candle_extended` augment the corresponding raw sets with technical indicators including SMA, RSI, MACD line and histogram, Williams %R, stochastic oscillator, and MFI. All features are lagged so that only information available at or before the current hour is used when generating a trade for the next-hour open entry and the following-hour open exit.

3.4 Train / Validation / Test Split

Each asset is divided chronologically into `50%` training, `25%` validation, and `25%` testing. Under the main 2021–2025 notebook logic this yields `21,902` training rows, `10,950` validation rows, and `10,952` test rows per asset after forward-return alignment and split-boundary trimming. The same protocol is reapplied to the shifted 2020–2024 rerun so that the robustness comparison changes the window, not the design. The evidence should therefore be read as two fixed-window out-of-sample evaluations, not as proof that the same rankings would persist under walk-forward re-estimation.

4 Methodology and Model Explanation

4.1 Prediction Labelling

The predictive task is framed as two binary classification problems built on the next-hour open-to-open log return. The strategy observes the current feature vector, enters at the next hours open if a threshold is triggered, and exits one hour later at the following open. We define:

- `label_up = 1` when the next-hour open-to-open log return is positive,
- `label_down = 1` when the next-hour open-to-open log return is negative.

This dual-labelling design allows long-only, short-only, and long-short trading structures to be implemented consistently across both machine-learning and rule-based strategies.

Class imbalance is limited in the submitted chronological split. Across the four training sets, the share of `label_up = 1` ranges from `49.2%` to `50.6%`, while the share of `label_down = 1` ranges from `48.3%` to `49.6%`; the residual corresponds to zero-return hours. Severe one-sided label dominance is therefore not the main driver of the reported results.

4.2 Strategy Families

4.2.1 Buy-and-Hold Benchmark

Buy-and-hold maintains a constant long position over the full test period and serves as the passive benchmark. In the `with_cost` setting, the benchmark incurs the one-time initial entry cost implied by moving from zero exposure to a fully invested long position.

4.2.2 Rule-Based Technical Heuristics

SMA and RSI serve as rule-based benchmarks. For SMA, short windows of `5`, `10`, `20`, and `30` hours are paired with long windows of `50`, `100`, `150`, and `200` hours. For RSI, windows of `7`, `14`, and `21` hours are tested. Each rule produces a continuous score, which is then mapped through the same validation and thresholding pipeline used for the machine-learning strategies. The long-only, short-only, and long-short structures therefore apply to SMA and RSI as well, rather than only to the ML models.

4.2.3 Machine-Learning Models

Four machine-learning models are evaluated. Logistic Regression provides a linear probability baseline. XGBoost and LightGBM represent tree-based ensembles that capture non-linear feature interactions. GRU is the sequential model, using a 24-hour lookback window to learn temporal dependencies in the input features. The GRU architecture used in the notebook is a compact recurrent

model with a 16-unit GRU layer followed by a dense layer, dropout, and a sigmoid output.

4.3 Backtesting Framework

Backtests are executed under both `no_cost` and `with_cost` regimes. The `with_cost` setting applies a transaction cost of `0.035%` per unit position change. Opening or closing a unit position therefore costs `0.035%`, while directly flipping from long to short costs `0.07%`. This design penalizes high-turnover strategies and makes it possible to separate raw predictive opportunity from economic viability after friction.

All portfolios are evaluated from a normalized initial capital of `1`. Positions are unit-scaled (`+1` long, `0` flat, `-1` short), so capital utilisation changes through time-in-market rather than through leverage sizing; this is why turnover and trading frequency are reported alongside the return-based metrics.

4.4 Evaluation Metrics and Feasibility Rules

Following the reference paper, the main benchmark is built from compounded signal returns rather than annualized hourly returns. Using the paper's notation,

$$\text{Total Return} = \prod_{i=1}^{N}(1 + R_i S_i)$$

$$\text{Average Return} = \frac{\text{Total Return}}{\sum_{i=1}^{N} S_i}$$

$$\text{Sharpe Ratio} = \frac{\text{Average Return}}{\text{Standard Deviation of Return}}$$

where R_i is the realized one-period return and $S_i \in \{0, 1\}$ indicates whether a signal is active at time i. In the notebook, this is implemented on the realized active net-return sequence r_j by using $(\prod_{j=1}^{N_{\text{sig}}}(1+r_j)-1)/N_{\text{sig}}$ in the numerator and dividing by the standard deviation of the signal-return sequence. This remains the headline Sharpe metric in the summary tables.

We also report annualized Sharpe as a secondary diagnostic:

$$\text{Annualized Sharpe} = \frac{\mu(r_t - r_f/8760)}{\sigma(r_t - r_f/8760)}\sqrt{8760}$$

where r_t denotes hourly net returns and $r_f = 0$ in the submitted implementation.

Thresholds are selected on the validation set using a precision-first rule subject to feasibility constraints. A candidate threshold must generate at least `20` trades, a signal ratio of at least `0.2%`, and recall of at least `1%` before it is considered feasible. Among feasible thresholds, precision is prioritized, with recall and validation Sharpe used only as secondary tie-breakers. The reports main tables, grouped statistical tests, and representative graphs focus on these feasible configurations, while infeasible cases are retained separately in diagnostic exports.

5 Model Performance Evaluation and Analysis

5.1 Overall Performance Comparison

Appendix A contains the supporting tables for the canonical 2021–2025 run, while Appendix C contains the window-comparison tables and figures for the additional 2020–2024 rerun. Table 1 shows the main-window ranking clearly: long-only is the strongest active structure, with average feasible test Sharpe `0.0495` in the no-cost case and `0.0212` after costs, while long-short falls to `0.0006` after costs and short-only remains negative. Table 9 preserves the same ranking but at a higher performance level in the earlier window: long-only rises to `0.0767` in the no-cost case and `0.0484` with costs, and long-short improves to `0.0433` and `0.0171`. Short-only does not share that improvement, moving from `0.0005` to `-0.0036` in the no-cost case and remaining near `-0.024` after costs. The ranking is therefore stable, but the level of performance is clearly regime-sensitive.

Buy-and-hold also remains a difficult benchmark in both windows. Table 2 shows that in the canonical test set the benchmark annualized Sharpe ratios are `0.82` for BTC, `0.53` for ETH, `1.40` for XRP, and `0.25` for SOL. Table 11 then shows that passive exposure is itself much stronger in

the earlier window: BTC cumulative return rises from 0.4010 to 2.2999, ETH from 0.1673 to 0.9302, XRP from 2.0132 to 2.9675, and SOL from -0.1723 to 2.3422. The earlier window is therefore friendlier not only to active strategies but also to the passive benchmark, which is why better absolute ML performance does not automatically imply better relative performance.

5.2 Impact of Transaction Costs

The no_cost versus with_cost comparison shows that transaction costs are not a secondary detail; they are central to the ranking of strategies in both windows. Table 1 shows that in 2021–2025, long-only average cumulative return falls from 0.2397 to -0.1065 when costs are introduced. Appendix C shows the same mechanism in 2020–2024: long-only falls from 0.4604 to 0.0292, and long-short falls from 0.3851 to -0.2100. The earlier regime is more forgiving, but it does not remove the economic burden of repeated position changes.

The same point appears in the positive-Sharpe shares reported in Table 9. In the with-cost setting, the share of positive-Sharpe feasible configurations rises from 50.0% to 73.6% for long-only and from 26.4% to 47.2% for long-short when the window is moved earlier, yet short-only falls from 19.4% to 13.9%. More complex strategies can generate more signals, but signal volume is not automatically a virtue when each position change incurs a cost, and a friendlier regime does not rescue every structure equally.

5.3 Model-Level Comparison

At the family level, Table 3 and Table 10 show that ML strategies are somewhat stronger than the rule-based benchmarks, but the advantage is regime-dependent. In the canonical 2021–2025 window, the average feasible test Sharpe is 0.0275 for ML models versus -0.0012 for rule-based strategies in the no-cost case, and -0.0004 versus -0.0056 after costs. In the 2020–2024 rerun, the corresponding ML averages improve to 0.0435 and 0.0162, while the rule-based averages move only to 0.0011 and -0.0050. ML therefore benefits more from the earlier, more favorable window, especially after costs, but the advantage remains modest rather than decisive.

Within the ML group, the outputs show a split picture rather than a single dominant winner. Tables 4 and 5 show that the canonical all-asset long-only averages are led by GRU in both cost regimes, with Logistic Regression also strong and XGBoost weakest on average. The earlier window broadens the leadership set instead of confirming one universal winner: Table 13 shows GRU, LightGBM, and Logistic Regression splitting the best-overall slots evenly in 2020–2024, whereas the 2021–2025 run is dominated by GRU and Logistic Regression. Extra model complexity therefore does not guarantee more robust trading performance.

5.4 Feature Representation and Strategy Behaviour

Within the feasible ML configurations, Tables 6 and 7 show the main-window representation result clearly: standard `ohlc` inputs outperform `candle` inputs in the grouped all-asset long-only with-cost comparison (test Sharpe `0.0255` versus `0.0195`), while extended feature sets outperform raw feature sets much more clearly (`0.0472` versus `-0.0024`). The current notebook therefore does not support the idea that candlestick inputs are broadly superior, but it does support the view that the paper-aligned extended indicators can help in selected settings. Table 12 reinforces that conclusion in the earlier window: significant grouped bootstrap results increase from `11/90` to `23/90`, and several of the additional significant rows are `extended_vs_raw` wins. Table 14 similarly shifts the best-overall feature counts toward `candle_extended` and `ohlc_extended` only. Predictive improvements therefore become easier to detect in the earlier window, yet they remain economically conditional because richer features can still coincide with higher trading activity.

The behavior of short-only strategies deserves explicit discussion because the underperformance is one of the most stable findings in the project. In the canonical with-cost setting, short-only strategies have a negative average Sharpe for every asset, ranging from `-0.0146` in BTC to `-0.0345` in ETH. Table 9 shows that the earlier window does not change that diagnosis: the average no-cost short-only Sharpe slips from `0.0005` to `-0.0036`, and the with-cost average remains around `-0.024`. This appears to reflect two forces: crypto markets retain enough upside drift that a pure short stance is structurally disadvantaged, and short-only strategies still trade frequently—about `1,564` trades on average in the later window and `1,382` in the earlier one—so even modest predictive errors are

quickly magnified by repeated transaction costs.

5.5 Statistical Significance

The notebooks main statistical layer uses grouped block-bootstrap tests on feasible ML configurations rather than exhaustive pairwise model-versus-model comparisons. These grouped tests compare `candle` versus `ohlc` inputs, `extended` versus `raw` feature sets, and `GRU` versus the rest of the ML set across asset scopes, strategy modes, and cost regimes. Table 8 shows that in the canonical 2021–2025 run, statistical significance is concentrated in a narrow subset of tests rather than appearing broadly across the grid. At the all-asset level, extended features outperform raw features in the with-cost short-only and long-short comparisons, and the `GRU`-versus-rest contrast is significant in those same with-cost grouped tests.

Table 12 shows that the earlier window yields more statistically detectable structure, but not a different bottom line. Significant grouped bootstrap results increase from `11/90` to `23/90`, and significant positive-usefulness tests versus zero rise from `4/24` to `9/24`. That means the 2020–2024 rerun makes more configurations look viable in absolute terms, especially for long-only and extended-feature ML setups. However, the same table also shows that the number of best-overall configurations that significantly outperform buy-and-hold remains `0` in both windows. Even where some configurations show positive Sharpe ratios or cumulative returns, the evidence of persistent alpha therefore remains narrow and unstable.

5.6 Equity Curve Analysis

The equity-curve figures are presented in Appendix B, and the regime-comparison figures are collected in Appendix C. The main-window equity curves show the asset-by-asset shape of the canonical 2021–2025 backtests, while the Appendix C figures summarize the cross-window comparison in aggregate. The long-only curves in Appendix B.1 show the clearest active-trading potential: BTC and XRP remain dominated by buy-and-hold after costs, ETH narrows the gap in selected configurations, and SOL is more competitive but also more uneven. The long-short curves in Appendix B.2

are noisier in all four assets and visually reinforce the cost drag already shown in the tables. The short-only curves in Appendix B.3 are the clearest confirmation of persistent weakness, with several best-per-model paths decaying steadily after costs. The Appendix C figures show the same story in aggregate: Figure 13 captures the strategy-mode shift, Figure 14 the model-family shift, and Figure 15 the stronger passive benchmark in the earlier window.

6 Conclusion, Limitations, and Future Work

6.1 Conclusion

Within the canonical 2021–2025 evaluation window, the answer to our research question is: only to a limited extent, and not in any broad or robust sense. Machine-learning models can improve on simple rule-based strategies in selected settings, especially before costs, but they do not consistently or statistically reliably outperform buy-and-hold once transaction costs are applied.

The additional 2020–2024 robustness rerun changes the strength of the results more than their ranking. Long-only and long-short strategies—especially ML configurations—become materially stronger in the earlier window, and more grouped tests and usefulness tests turn significant. That confirms that the framework is regime-sensitive rather than uniformly weak. Even so, the broader conclusion does not flip. Short-only remains weak across both windows, transaction costs continue to erode long-short and high-turnover strategies, and none of the best configurations significantly outperform buy-and-hold in either sample. The evidence therefore supports a narrow claim: ML-based cryptocurrency strategies can outperform simple rules in selected settings, and they look better in friendlier regimes, but they do not deliver broad, cost-robust, statistically reliable superiority over passive buy-and-hold.

6.2 Limitations

Several limitations materially qualify the findings. First, even with the additional 2020–2024 rerun, the study still relies on fixed-window comparisons rather than rolling or expanding walk-forward

retraining. The second window improves the regime-sensitivity evidence, but it does not show whether the observed rankings would persist under continuously updated historical partitions.

Second, the predictive target is binary directional classification rather than return regression. A correct prediction on a very small move is therefore treated the same as a correct prediction on a much larger economically important move. The thresholding rules partially mitigate this by filtering for higher-confidence signals, but they do not remove the mismatch between directional classification and return magnitude.

Third, the backtest remains stylized. It applies exchange-style transaction fees, but it does not model slippage, execution delay, order-book depth, market impact, borrowing costs for shorting, margin requirements, or liquidation dynamics.

Fourth, the models trade at hourly frequency. This preserves an intraday research design and aligns with the reference paper, but it also increases exposure to noise and to turnover-driven cost drag.

Finally, the dataset is exchange-specific and limited to four assets, so some conclusions may nor reflect the full cryptocurrency universe. The current version also does not include a dedicated explainability analysis or formal feature-selection stage, so it does not isolate which indicators contribute most to predictive performance or test dimensionality-reduction alternatives systematically.

6.3 Future Work

The main contribution of the project is comparative and cautionary rather than celebratory. Under two fixed historical windows and a realistic fee model, the practical value of active cryptocurrency trading remains limited, and the margin between machine learning and simpler alternatives is much narrower than headline predictive results might suggest. Future work should move from fixed-window comparison to walk-forward re-estimation, add explicit regime labelling, incorporate richer backtest frictions, explore lower-frequency variants, and extend explainability analysis to determine whether any observed advantage survives under stronger robustness checks.

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A Appendix A: Tables

The full exported CSV tables remain available under `result/paper_tables/`. This appendix reproduces the most important report-facing tables so that the main body can stay interpretation-first.

Table 1: Average feasible test outcomes across strategy modes.

Strategy Mode	Test Sharpe (No Cost)	Test Sharpe (With Cost)	Cum. Return (No Cost)	Cum. Return (With Cost)	Number of Trades
Long-only	0.0495	0.0212	0.2397	-0.1065	1068
Long-short	0.0236	0.0006	0.1176	-0.4217	2144
Short-only	0.0005	-0.0242	-0.0217	-0.4057	1564

Note: All numeric entries are averages across feasible configurations.

Table 2: Buy-and-hold benchmark summary on the test set.

Asset	Cumulative Return	Annualized Return	Annualized Sharpe	Maximum Drawdown
BTC	0.4010	0.3095	0.8180	-0.3476
ETH	0.1673	0.1317	0.5303	-0.6528
XRP	2.0132	1.4163	1.4012	-0.5121
SOL	-0.1723	-0.1403	0.2523	-0.6606

Table 3: Average feasible test outcomes by model family.

Cost Regime	ML Sharpe	Rule Sharpe	ML Cum. Return	Rule Cum. Return
No Cost	0.0275	-0.0012	0.1369	-0.0887
With Cost	-0.0004	-0.0056	-0.3060	-0.3537

Note: All numeric entries are averages across feasible configurations.

Table 4: All-asset long-only model averages, no-cost feasible configurations.

Model	Test Sharpe	Test Precision	Test Recall	Test Avg. Return
GRU	0.0773	0.5885	0.0845	0.001050
Logistic Regression	0.0690	0.5582	0.0386	0.000825
LightGBM	0.0536	0.5481	0.0725	0.000653
SMA	0.0286	0.5239	0.0585	0.000379
XGBoost	0.0175	0.5176	0.1731	0.000140
RSI	0.0069	0.5335	0.4517	0.000069

Table 5: All-asset long-only model averages, with-cost feasible configurations.

Model	Test Sharpe	Test Precision	Test Recall	Test Avg. Return
GRU	0.0530	0.5885	0.0845	0.000846
Logistic Regression	0.0382	0.5582	0.0386	0.000550
SMA	0.0266	0.5239	0.0585	0.000362
LightGBM	0.0151	0.5481	0.0725	0.000379
RSI	-0.0026	0.5335	0.4517	0.000001
XGBoost	-0.0128	0.5176	0.1731	-0.000098

Table 6: All-asset long-only input-type comparison, with-cost feasible configurations.

Input Type	Test Sharpe	Test Precision	Test Recall	Test Avg. Return
ohlc	0.0255	0.5629	0.0855	0.000425
candle	0.0195	0.5425	0.0989	0.000401

Table 7: All-asset long-only feature-type comparison, with-cost feasible configurations.

Feature Type	Test Sharpe	Test Precision	Test Recall	Test Avg. Return
extended	0.0472	0.5808	0.0579	0.000809
raw	-0.0024	0.5234	0.1264	0.000003

Table 8: Significant grouped bootstrap results from the feasible configuration set.

Scope	Strategy Mode	Cost Regime	Test Type	Mean Diff	p-value
all_assets	short_only	with_cost	extended_vs_raw	0.000027	0.0379

Scope	Strategy Mode	Cost Regime	Test Type	Mean Diff	p-value
all_assets	short_only	with_cost	gru_vs_other_models	0.000026	0.0219
all_assets	long_short	with_cost	extended_vs_raw	0.000037	0.0112
all_assets	long_short	with_cost	gru_vs_other_models	0.000062	0.0000
btc	long_only	with_cost	gru_vs_other_models	0.000027	0.0002
btc	short_only	with_cost	extended_vs_raw	0.000040	0.0005
btc	short_only	with_cost	gru_vs_other_models	0.000049	0.0000
btc	long_short	with_cost	gru_vs_other_models	0.000068	0.0000
eth	long_short	with_cost	extended_vs_raw	0.000036	0.0420
eth	long_short	with_cost	gru_vs_other_models	0.000068	0.0000
xrp	long_only	no_cost	gru_vs_other_models	-0.000021	0.0236

B Appendix B: Figures

B.1 Long-Only With-Cost Equity Curves

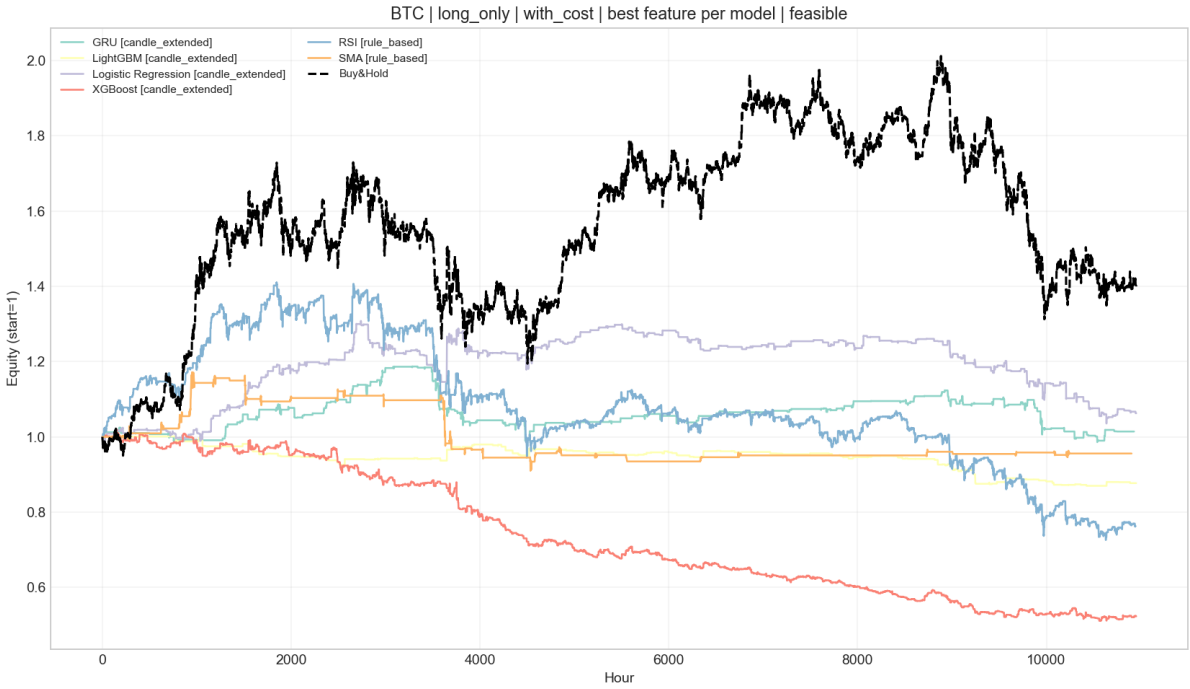

Figure 1: BTC long-only equity curves under transaction costs.

Figure 2: ETH long-only equity curves under transaction costs.

Figure 3: SOL long-only equity curves under transaction costs.

Figure 4: XRP long-only equity curves under transaction costs.

B.2 Long-Short With-Cost Equity Curves

Figure 5: BTC long-short equity curves under transaction costs.

Figure 6: ETH long-short equity curves under transaction costs.

Figure 7: SOL long-short equity curves under transaction costs.

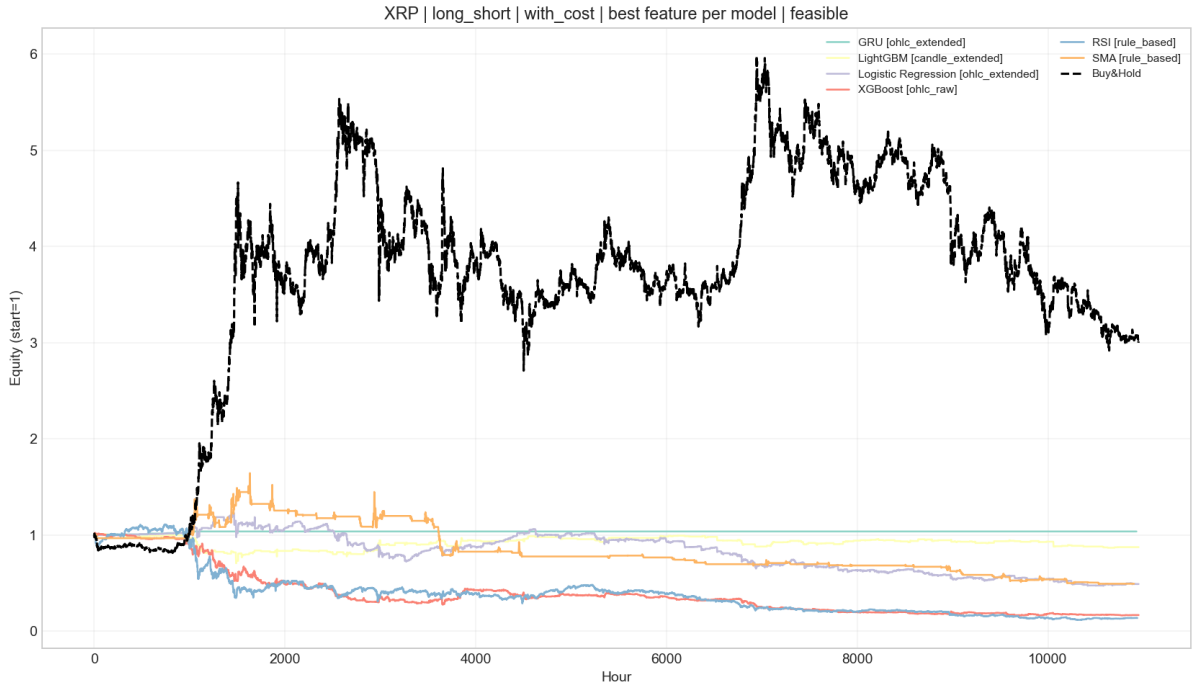

Figure 8: XRP long-short equity curves under transaction costs.

B.3 Short-Only With-Cost Equity Curves

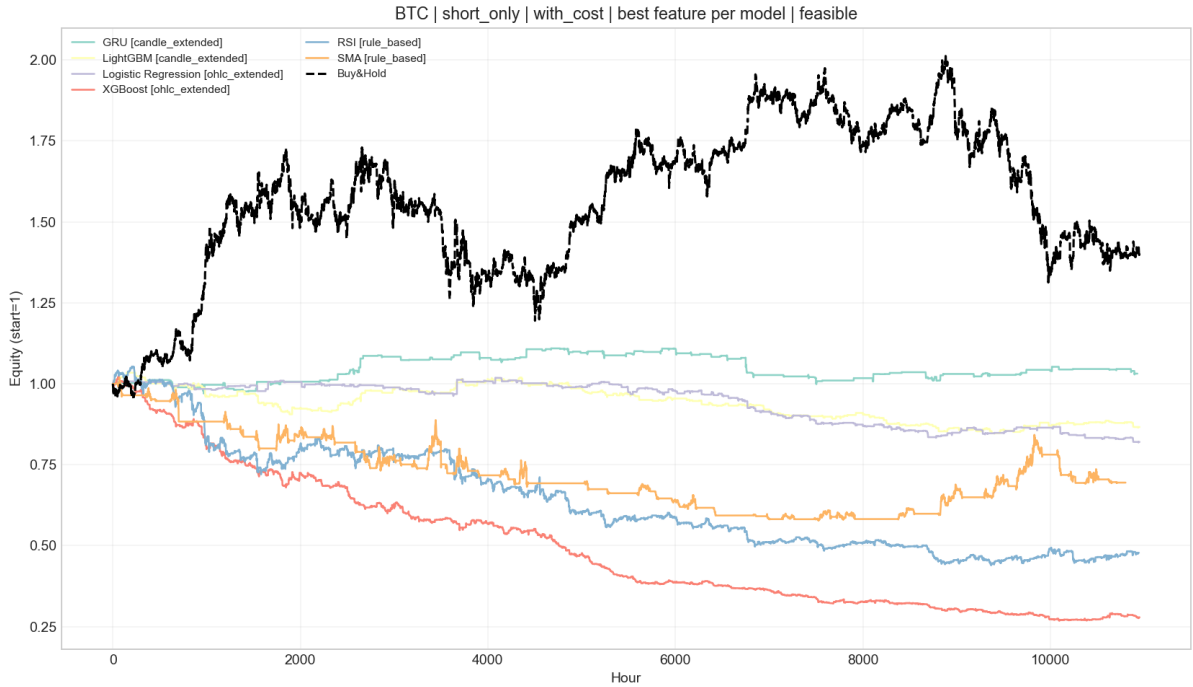

Figure 9: BTC short-only equity curves under transaction costs.

Figure 10: ETH short-only equity curves under transaction costs.

Figure 11: SOL short-only equity curves under transaction costs.

Figure 12: XRP short-only equity curves under transaction costs.

C Appendix C: Regime Robustness Comparison

C.1 Window-Comparison Tables

Table 9: Strategy-mode comparison across the 2021–2025 and 2020–2024 windows.

Window	Strategy	Cost	Sharpe	Cum. Return	Trades	Pos. Sharpe
2021–2025	Long-only	No Cost	0.0495	0.2397	1068	79.2%
2021–2025	Long-only	With Cost	0.0212	-0.1065	1068	50.0%
2021–2025	Long-short	No Cost	0.0236	0.1176	2144	62.5%
2021–2025	Long-short	With Cost	0.0006	-0.4217	2144	26.4%
2021–2025	Short-only	No Cost	0.0005	-0.0217	1564	45.8%
2021–2025	Short-only	With Cost	-0.0242	-0.4057	1564	19.4%
2020–2024	Long-only	No Cost	0.0767	0.4604	1119	90.3%
2020–2024	Long-only	With Cost	0.0484	0.0292	1119	73.6%
2020–2024	Long-short	No Cost	0.0433	0.3851	1908	77.8%
2020–2024	Long-short	With Cost	0.0171	-0.2100	1908	47.2%
2020–2024	Short-only	No Cost	-0.0036	-0.1405	1382	40.3%
2020–2024	Short-only	With Cost	-0.0241	-0.4324	1382	13.9%

Note: Sharpe, cumulative return, and trade columns are averages across feasible configurations. Positive Sharpe reports the share of feasible configurations with positive test Sharpe.

Table 10: Model-family comparison across the 2021–2025 and 2020–2024 windows.

Window	Family	Cost	Sharpe	Cum. Return	Trades
2021–2025	ML	No Cost	0.0275	0.1369	1656
2021–2025	ML	With Cost	-0.0004	-0.3060	1656
2021–2025	Rule	No Cost	-0.0012	-0.0887	1077
2021–2025	Rule	With Cost	-0.0056	-0.3537	1077
2020–2024	ML	No Cost	0.0435	0.2402	1520
2020–2024	ML	With Cost	0.0162	-0.2091	1520
2020–2024	Rule	No Cost	0.0011	0.1932	1062
2020–2024	Rule	With Cost	-0.0050	-0.1671	1062

Note: All numeric entries are averages across feasible configurations.

Table 11: Buy-and-hold comparison across the two evaluation windows.

Window	Asset	Cum. Return	Ann. Return	Ann. Sharpe	Max Drawdown
2021–2025	BTC	0.4010	0.3095	0.8180	-0.3476
2021–2025	ETH	0.1673	0.1317	0.5303	-0.6528
2021–2025	XRP	2.0132	1.4163	1.4012	-0.5121
2021–2025	SOL	-0.1723	-0.1403	0.2523	-0.6606
2020–2024	BTC	2.2999	1.5981	2.1472	-0.3231
2020–2024	ETH	0.9302	0.6920	1.1677	-0.4666
2020–2024	XRP	2.9675	2.0105	1.7153	-0.4578
2020–2024	SOL	2.3422	2.0011	1.6357	-0.4652

Table 12: Significance summary across the two evaluation windows.

Window	Grouped Sig.	Grouped Total	Useful vs 0	Useful Total	Beat B&H Sig.
2021–2025	11	90	4	24	0
2020–2024	23	90	9	24	0

Table 13: Best-overall model counts across the two evaluation windows.

Window	Model	Count
2021–2025	GRU	12
2021–2025	Logistic Regression	12
2020–2024	GRU	8
2020–2024	LightGBM	8
2020–2024	Logistic Regression	8

Table 14: Best-overall feature-set counts across the two evaluation windows.

Window	Feature Set	Count
2021–2025	ohlc_extended	12
2021–2025	candle_extended	10
2021–2025	ohlc_raw	2
2020–2024	candle_extended	14
2020–2024	ohlc_extended	10

C.2 Window-Comparison Figures

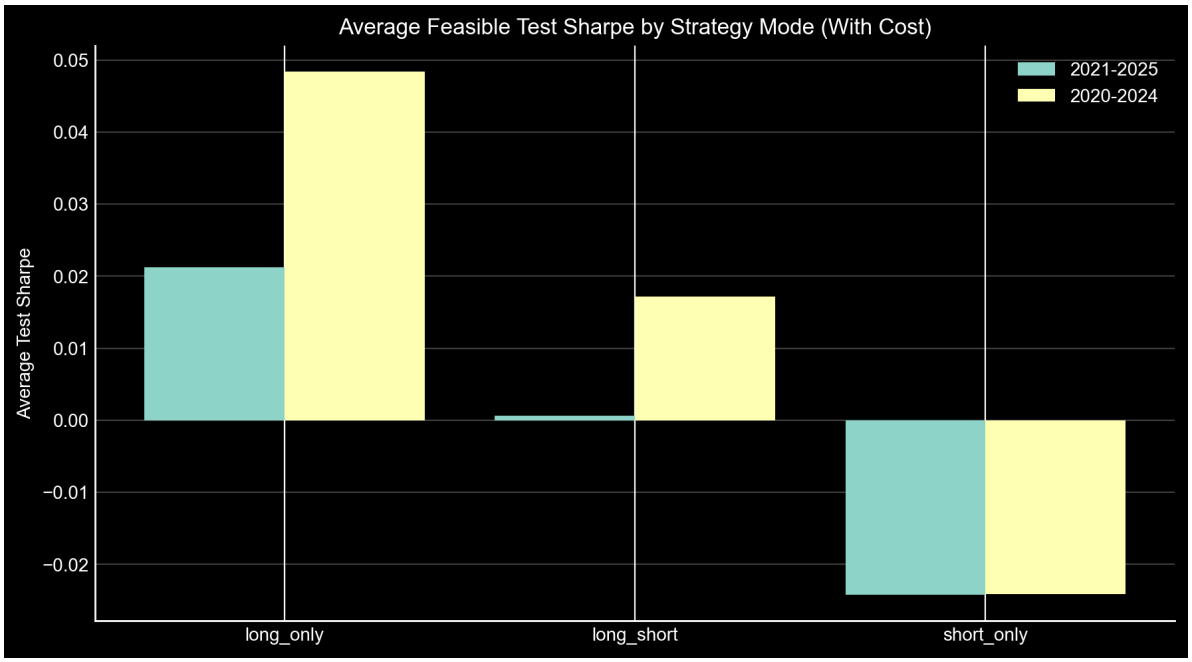

Figure 13: Average feasible test Sharpe by strategy mode, with cost, across the two windows.

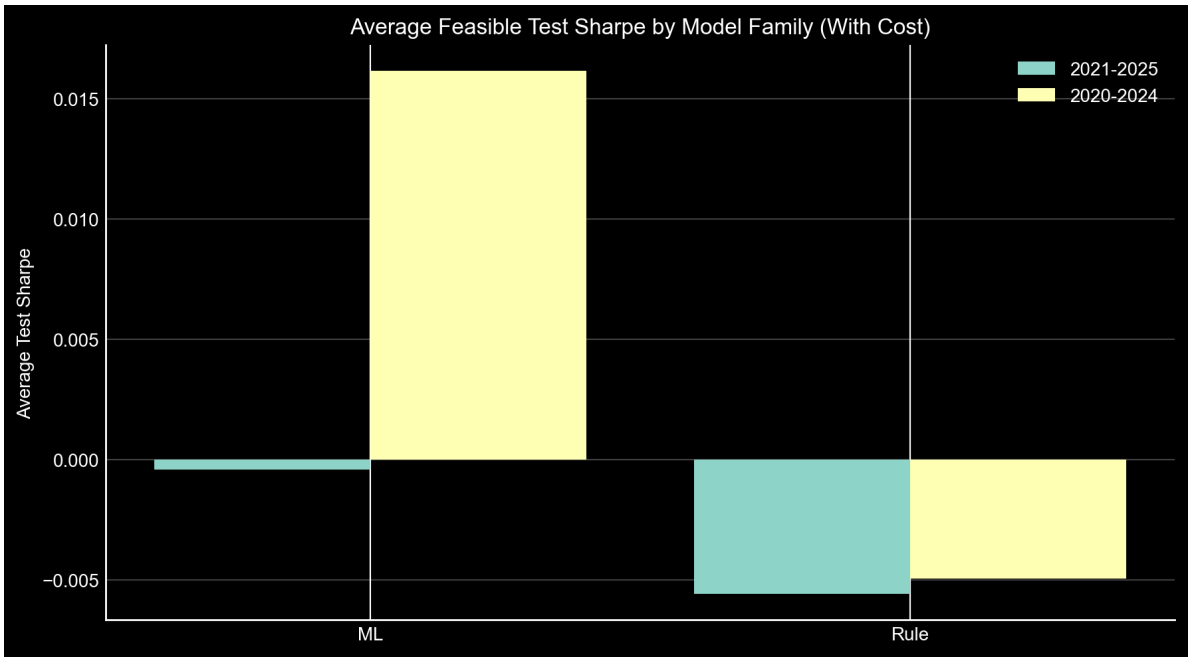

Figure 14: Average feasible test Sharpe by model family, with cost, across the two windows.

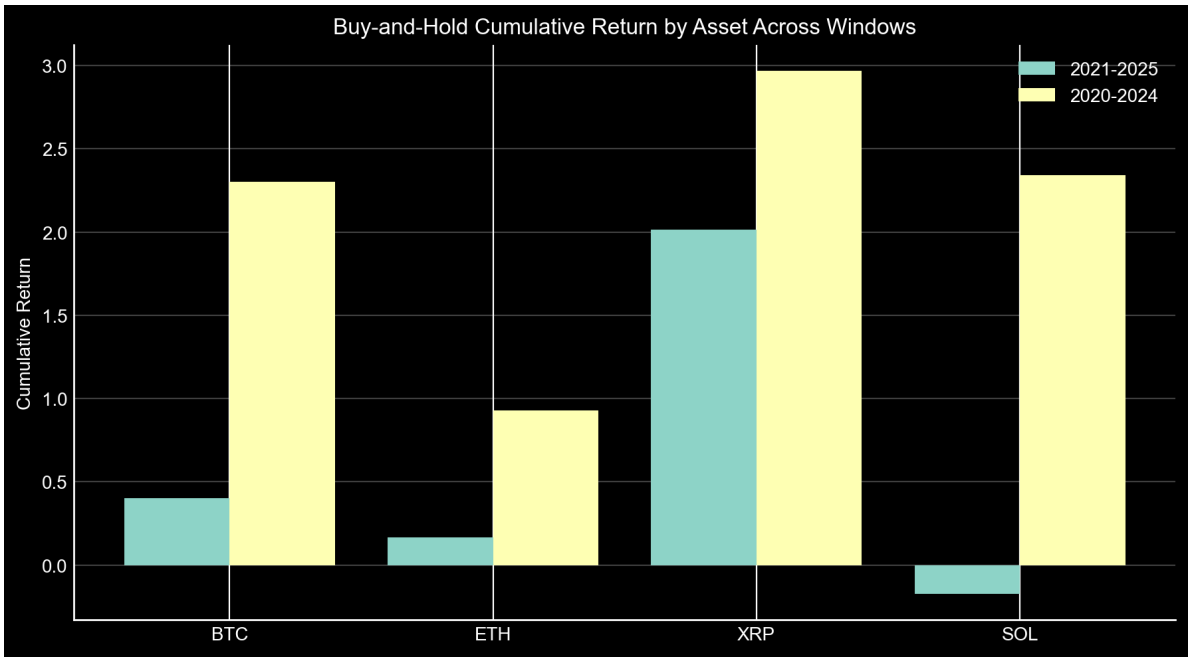

Figure 15: Buy-and-hold cumulative return by asset across the two windows.

Supplementary Rebuttal

We summarized the suggestions from the five classmates—1: `3dkdiz`, 2: `3eyyuc`, 3: `3radnt`, 4: `3sgrqy`, and 5: `3sskvf`—into 28 concise points below.

1. **Single train/test split**: One fixed historical window may not generalise across market regimes. (by 2, 4, 5) *Response:* this remains a real limitation, but it is now partially mitigated by the additional 2020–2024 fixed-window robustness comparison in Section 5 and Appendix C. That second window shows regime sensitivity directly, even though it does not replace walk-forward validation.

2. **Regime-conditioned analysis**: Reporting results by regime would strengthen the generalisation claim. (by 2, 3) *Response:* partially addressed via the additional 2020–2024 rerun, which keeps the same pipeline and tests whether the conclusions change in an earlier market environment. Formal regime labels and within-window regime segmentation remain future work.

3. **Binary classification target**: Direction-only prediction ignores return magnitude. (by 2, 3, 4) *Response:* we kept the binary target to stay aligned with the core project design and to preserve a common thresholding framework across ML and rule-based models. The mismatch with economic payoff is now stated explicitly in *Limitations*.

4. **Sharpe formulation and code consistency**: Sharpe definitions were unclear. (mainly by 2) *Response:* fixed in *Evaluation Metrics and Feasibility Rules*, which now states the paper-style Sharpe formula explicitly, explains the notebook implementation, and separates it from annualized Sharpe. The report and code are now aligned on the same primary definition.

5. **Short-only underperformance unexplained**: The consistent weakness of short-only strategies needed interpretation. (by 1) *Response:* fixed in *Feature Representation and Strategy Behaviour*. The report now links short-only weakness to persistent upside drift in crypto and to the heavier fee burden from repeated position changes.

6. **Tuning asymmetry**: Reviewers thought ML was tuned while SMA/RSI were not. (by 1) *Response:* rebutted in *Rule-Based Technical Heuristics*. SMA and RSI also go through parameter search and then enter the same validation-threshold pipeline, so the asymmetry was more apparent in the writeup than in the code.

7. **Transaction-cost sensitivity**: A broader fee sweep was suggested. (by 1, 5) *Response:* we keep the Binance-aligned fee design in the submitted study. The `0.035%` rate is the midpoint of Binance USDT maker and taker fees (`0.0200%`/`0.0500%`), corresponding to a simple assumption that fills are maker-like about half the time and taker-like about half the time; a broader fee sweep is left for future work.

8. **Backtest realism**: Slippage, liquidity, and realistic shorting frictions are omitted. (by 3, 4) *Response:* admitted directly in *Limitations*. The submitted backtest is fee-aware rather than fully execution-realistic, so the results should be read as stricter than a frictionless backtest but still simpler than live trading.

9. **Precision-first tuning vs Sharpe objective**: The selection objective looked mismatched with the economic objective. (by 4) *Response:* clarified in *Evaluation Metrics and Feasibility Rules* as a deliberate high-confidence signal design. Precision is used to filter noisy trades first, while Sharpe remains the main reporting metric and a secondary tie-breaker during threshold choice.

10. **Noisy hourly frequency**: Hourly trading may be noise-heavy and turnover-heavy. (by 3) *Response:* acknowledged in *Sample Period* and *Limitations*. We kept the hourly horizon for paper alignment and to retain an intraday setting, but the report now states clearly that this choice increases exposure to noise and fee drag.

11. **Cluttered equity curves**: The original bundled plots were too crowded. (by 1) *Response:* fixed by moving the figure set to the appendix and using representative best-per-model curves instead of dumping every line into the main body. Section 5.6 now explains what each figure group is meant to show.

12. **Conclusion did not answer the question directly**: The central conclusion was too indirect. (by 1, 5) *Response:* fixed in *Conclusion*. The final section now answers the stated research question directly and then qualifies that answer with the cost and regime-robustness evidence.

13. **Feature importance analysis absent**: The report did not explain which indicators drive the extended-feature results. (by 1, 3) *Response:* we did not add SHAP-style or feature-importance analysis in this round. Instead, the report now states this gap more clearly under explainability and feature-selection limitations.

14. **Rule-based strategy implementation vague**: It was unclear whether long-only, short-only, and long-short applied to SMA/RSI too. (by 5) *Response:* fixed in *Rule-Based Technical Heuristics*. The report now states explicitly that the three strategy structures apply to SMA and RSI as well, not only to the ML models.

15. **Infeasible configurations not explained**: Excluded configurations needed a principled explanation. (by 5) *Response:* fixed in *Evaluation Metrics and Feasibility Rules*. The report now explains that infeasibility is driven by minimum trades, signal-ratio, and recall thresholds rather than by discretionary exclusion.

16. **Literature review too long**: More space should be devoted to interpretation and methodology. (by 5) *Response:* fixed by compressing the literature review into three transition-linked paragraphs and shifting more space toward Section 5. The review keeps the core named studies, but no longer crowds out the empirical discussion.

17. **No README**: The repo previously lacked a clear entry point. (by 1, 2, 3, 4, 5) *Response:* fixed in `README.md`, which now explains the repo structure, the role of `project.ipynb`, the optional sandbox, and a typical grader workflow. This should make the repo readable without guessing the pipeline order.

18. **Inaccessible GitHub repository**: The original review link was not publicly accessible. (by 1, 2) *Response:* this is an operational issue rather than a methodological one, but it still matters. The repo has been cleaned for a public handoff, and the final submission should use an accessible link rather than the earlier private state.

19. **Redundant notebooks**: Multiple notebooks created ambiguity about the canonical pipeline. (by 1) *Response:* fixed by consolidating the repo around `project.ipynb` as the canonical notebook. Older exploratory notebooks are no longer part of the tracked core.

20. **GRU undocumented**: The most complex model needed clearer documentation. (by 1) *Response:* fixed in *Machine-Learning Models* and the canonical notebook. The architecture, lookback window, and training role of the GRU are now described in the report instead of being left implicit.

21. **Messy results output**: The result folders were hard to navigate. (by 4) *Response:* partially fixed by cleaning the repo structure, surfacing the report-facing outputs, and reducing tracked generated files. The result directories are still detailed, but the important tables and figures are now easier to identify.

22. **Reproducibility failure (GRU)**: Earlier reruns produced different GRU results. (by 4) *Response:* mitigated by global seeding and deterministic TensorFlow settings in `project.ipynb`. After that, the final reported values were resynchronised with the rerun outputs so the report reflects the current pipeline rather than stale results.

23. **No requirements file**: Environment setup was underspecified. (by 1) *Response:* fixed in `requirements.txt`. The dependency list is now grouped and more explicit about the notebook, modelling, and TensorFlow stack used in the project.

24. **No cross-validation**: Rolling or walk-forward validation would provide stronger evidence of stability. (by 3, 5) *Response:* still admitted as a limitation in *Limitations*. The added 2020–2024 rerun strengthens the robustness discussion, but it does not replace rolling or walk-forward retraining evidence.

25. **Class imbalance not addressed**: The report should show whether the labels are materially imbalanced. (by 3) *Response:* fixed with the class-balance note in *Prediction Labelling*. The report now states that the training labels are close to balanced, so one-sided class dominance is not the main explanation for the results.

26. **Limited hyperparameter tuning for ML models**: The ML models rely on mostly fixed hyperparameters. (by 3) *Response:* we did not expand ML hyperparameter search in this round. The submitted design keeps mostly fixed model settings for comparability and scope control, and this remains a limitation of the final study.

27. **No feature selection or dimensionality reduction**: Redundancy and multicollinearity may remain in the engineered features. (by 3, 5) *Response:* admitted as a limitation in *Limitations*. We kept the paper-aligned feature families rather than adding a second modelling layer for selection or PCA, but the report now states this tradeoff more plainly.

28. **No model explainability**: The study does not yet show which features drive predictions. (by 3) *Response:* admitted as a limitation in *Limitations* and *Future Work*. The report now makes clear that the current contribution is comparative and backtest-oriented rather than an explainability study.